Water Quality C - Water Quality - University of Chicago C - 01-16-2021

There are three main sections, corresponding with those outlined in the rules. Because of this digital format, the salinometer portion will not be conducted.

Part 1 Questions: Marine and Estuary Ecology

1. (1.00 pts) What are the six aquatic ecosystems?

Expected Answer: Oceans, lakes, rivers, streams, estuaries, and wetlands

2. (1.00 pts) How does water return to the ground in the water cycle?

- A) Evaporation
- B) Sublimation
- C) Precipitation
- D) Condensation

3. (1.00 pts) Which of the following could affect infiltration of water into the surface of the soil?

I. Concrete sidewalks

II. Roads

III. Parks and gardens

- A) III only
- B) I and II
- C) I only
- D) II and III

4. (1.00 pts) What is transpiration?

Expected Answer: When plants lose water through its leaves, it is called transpiration

5. (1.00 pts) Describe nutrient cycling.

Expected Answer: A nutrient cycle refers to the movement and exchange of organic and inorganic matter back into the production of living matter.

6. (1.00 pts) What source of energy drives the water cycle?

- A) Magma
- B) The Sun
- C) Gravity
- D) Nuclear

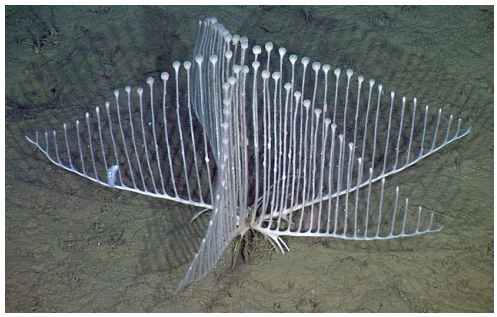
7. (1.00 pts) In a nutrient cycle, nutrients are:

- A) Recycled constantly
- B) Put in long term storage occasionally in nutrient "sinks"
- C) Put in the bodies of organisms temporarily
- D) All of the above

8. (4.00 pts) What conditions allow for the continued existence of the first reef-building organisms?

- A) Warm, shallow water
- B) Saline Lagoons
- C) Cool, shallow water
- D) Deep, anoxic water
- E) They went extinct

9. (1.00 pts) What differentiates this from its shallower relatives?



- A) Shape; relatives form denser calcareous masses
- B) Size; this organism is much larger than its more characteristic relatives
- C) Diet; this organism is carnivorous
- D) Reproduction; this organism reproduces sexually rather than asexually
- E) Lifespan; this organism only lives for a maximum 50 years

10. (3.00 pts) Identify the specimen this came from:



Expected Answer: Tridacna (spp.)

11. (1.00 pts) In what way do juvenile Sweetlips *not* resemble adult ones? Select all that apply.

(Mark **ALL** correct answers)

- [ ] A) Eat other fish
- [x] B) Live solitary lives
- [x] C) Different coloration
- [ ] D) Different habitats
- [x] E) Different swimming style

12. (2.00 pts) Which Atlantic ocean dwellers work together (not willingly) for one's organism's reproduction?

- ( ) A) Gorgonia and the Flamingo Tongue Snail. The snail will burrow into the sediment near the coral's base, providing the snail young a food source as they mature
- ( ) B) Nassau Grouper and Gorgonia. When searching for spawning grounds, groupers have been observed to follow specific Gorgonias as cues for what direction to take
- ( ) C) Pencil Urchin and Moray Eel. Moray Eel young will use these urchins as temporary habitats when searching for a proper home
- (•) D) Gorgonia and the Flamingo Tongue Snail. Female snails will lay their eggs in the Gorgonia's axis where feeding has previously taken place
- ( ) E) Nassau Grouper and Gorgonia. Gorgonias will often capture eggs that are also quickly fertilized in the coral's branches. This also increases survival rate

13. (1.00 pts) What survival curve do humans follow?

- (•) A) Type I
- ( ) B) Type II
- ( ) C) III

14. (1.00 pts) What survival curve do flies use?

- ( ) A) Type I
- ( ) B) Type II
- (•) C) Type III

15. (1.00 pts) An r strategy is associated with which of the following characteristics? Please choose all that apply.

(Mark **ALL** correct answers)

- [x] A) Small size
- [ ] B) A lot of energy is expended in order to produce and raise each individual
- [ ] C) Long life expectancy
- [x] D) Each member of the population reproduces a single time
- [ ] E) Type I survival curve
- [x] F) Produces many offspring, not all of which survive

16. (1.00 pts) What is another name for an algal bloom?

- ( ) A) Protozoan infection
- (•) B) Red tide
- ( ) C) Algae invasion
- ( ) D) Marine invasion

17. (1.00 pts) Lake Erie frequently suffers from severe algal blooms. Runoff of which of the following substances is the main cause?

- ( ) A) Pesticides
- ( ) B) Food waste
- ( ) C) Industrial waste
- (•) D) Fertilizers

18. (1.00 pts)
A volcano has erupted in the Pacific Ocean, leading to the formation of new rock. Coral begins to grow on this surface. Which term accurately describes this process?

- (•) A) Primary Succession
- ( ) B) Secondary Succession
- ( ) C) Tertiary Succession
- ( ) D) Quaternary Succession
- ( ) E) Primary Colonization

19. (1.00 pts)
The Chinese Mitten Crab is an invasive species that has recently become a serious issue in the estuary of the Hudson River. Which reason below is not a valid reason that these crabs may spread dangerously in the ecosystem?

- ( ) A) Rapid spread due to lack of predators
- ( ) B) Ability to outcompete native species
- ( ) C) Lasting damage to environment due to changes in the environment
- (•) D) Cross breeding with native species

20. (1.00 pts) Which step of the wastewater treatment process involves using filters to remove solid waste and allowing water to sit and solids to settle at the bottom (sludge)?

- (●) A) Primary
- ( ) B) Secondary
- ( ) C) Tertiary
- ( ) D) Quaternary

21. (1.00 pts) Please select the levels of treatment most common in (used in pretty much all) municipal waste water treatment plants.

(Mark **ALL** correct answers)

- [x] A) Primary
- [x] B) Secondary
- [ ] C) Tertiary
- [ ] D) Quaternary

22. (4.00 pts)
What is a saprobic index? What information do they provide, and what are the assumptions involved when using them? What are the four zones of gradual self-purification?

Expected Answer: Objective: - aims to provide a water quality classification by means of a system of aquatic organisms Principle: - every species has a specific dependency of decomposing organic substances and thus the oxygen content: this tolerance is expressed in a saprobic indicator value Advantages: - quick classification of the investigated community (saprobic index) can be made on a universal scale - saprobic index can be obtained for several biotic groups Problems: - identification of organisms at species level required - saprobic index calculation requires assessment of abundance - the saprobic system implies more knowledge than actually exists: pollution tolerances are highly subjective and based on ecological observations and rarely confirmed by experimental studies Four zones of gradual self-purification: the polysaprobic zone, the α-mesosaprobic zone, the β-mesosaprobic zone, and the oligosaprobic zone

23. (1.00 pts) Why are microplastic pollutants dangerous to marine life?

Expected Answer: bioaccumulation of toxic chemicals in digestive tracts and gills (EDCs endocrine disruptor chemicals)

24. (1.00 pts) Give 1 example of a fish species (common name) for each type of migration behavior: anadromous, catadromous.

Expected Answer: anadromous: salmon, lamprey, sturgeon, herrings, basses, smelts catadromous: freshwater eel

25. (1.00 pts) What type of locomotion does the tuna fish use?

- ( ) A) carangiform
- (●) B) thunniform

C) anguilliform
D) ballistiform

26. (1.00 pts) Corals and zooxanthellae have what type of interspecies interaction?

A) Commensalism
B) Predation
C) Parasitism
D) Mutualism
E) Neutralism

27. (1.00 pts) Aragonite saturation is used to track the calcium ion concentration in oceans.

True False

28. (1.00 pts) Tritons (genus *Charonia)* are hermaphrodites and undergo self fertilization.

True False

29. (1.00 pts) What is the function of the coenosarc in corals?

Expected Answer: The living tissue overlying the stony skeletal material of the coral. It secretes the coenosteum, the layer of skeletal material lying between the corallites (the stony cups in which the polyps sit). The coenosarc contains the gastrovascular canal system that links the polyps and allow them to share nutrients and symbiotic zooxanthellae

30. (1.00 pts) What type of toxins are present in the spines of the crown of thorns starfish?

A) Neurotoxin
B) Myotoxin
C) Necrotoxin
D) Saponin
E) Cyanotoxin

Congratulations! You have finished the test. Be sure to double check your answers.

Good luck in the tournament!

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